

# John Che

*PhD Student · Computer Science · Human-Centered AI & Recommender Systems*

University of Minnesota Twin Cities · Minneapolis, MN

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## RESEARCH FOCUS

PhD student at the GroupLens Research Lab interested in human-centered recommender systems, algorithmic bias, and information literacy in personalized news environments. Prior work spans passkey security, multilingual NLP, and large-scale security data pipelines. Interested in research collaborations in any of these areas.

## EDUCATION

### University of Minnesota Twin Cities – Expected 2031

*PhD, Computer Science · Machine Learning & Recommender Systems · Minneapolis, MN*

- Advisors: Dr. Joseph Konstan & Dr. Loren Terveen · GroupLens Research Lab
- Research area: human-centered news recommender systems

### University of Wisconsin Madison – 2025

*MSc, Computer Science · Machine Learning & Data Engineering · Madison, WI*

- Graduate Research Fellow, UW System Office of Information Security (UWSA Fellowship, 1 of 7 statewide)
- Research advisors: Dr. Rahul Chatterjee (Security & Privacy); Dr. Shivaram Venkataraman (ML Systems)

### University of Wisconsin Madison – 2023

*BSc, Computer Engineering · Summa Cum Laude · Dean's Honor List, 7 semesters · Madison, WI*

- King-Morgridge Scholarship (2019–2023) — full-tuition merit award
- ASCAM Leadership Excellence Award (2023)

## RESEARCH EXPERIENCE

### Graduate Researcher Aug. 2026 - Present

*Advisors: Dr. Joseph Konstan & Dr. Loren Terveen · GroupLens Research Lab · University of Minnesota - Twin Cities, Minnesota*

- Conducting foundational research in social computing, recommender systems, and HCI; gaining hands-on experience with the full research lifecycle, from problem formulation and study design to data analysis and academic writing; investigating algorithms and interfaces for systems like MovieLens and the LensKit open-source toolkit

### Graduate Researcher Sept. 2024 - Dec. 2024

*Dr. Rahul Chatterjee · University of Wisconsin Madison · Madison, WI*

- Designed a full IRB-approved qualitative study on passkey adoption barriers: recruitment, semi-structured interviews (n=12), thematic coding
- Developed a taxonomy of “bootstrap friction points” in passwordless authentication identifying cross-ecosystem incompatibility and poor documentation as primary blockers
- Surfaced user misconceptions about biometric authentication privacy; manuscript under review at CHI 2026 and USENIX Security

### Graduate Researcher Jan. 2024 - May 2024

*Dr. Shivaram Venkataraman · University of Wisconsin Madison · Madison, WI*

- Implemented distributed training pipeline (PyTorch DistributedDataParallel on Google Cloud) for multilingual mT5-small (300M parameters)
- ↑ **15% BLEU** · **1/7th cost**: Achieved 15% higher BLEU scores at one-seventh the cost of an A100 baseline
- Presented findings at UW Madison Graduate Research Symposium (2024)

### Office of Information Security · Research Fellow Aug 2024 – May 2025

*UW System · UWSA Fellowship · Serves 165K+ users across 13 campuses · Madison, WI*

- ↓ **30% latency**: Designed A/B experiments that reduced threat-detection-to-mitigation latency by 30%
- Evaluated data pipeline architectures handling 5 TB/day of security events across 13 campus environments
- Built real-time event-streaming infrastructure using .NET, MS SQL, and Elasticsearch; presented findings to security leadership and policymakers

## Wisconsin Power Systems Lab · Undergraduate Research Assistant 2020 – 2021

University of Wisconsin Madison · Madison, WI

- Analyzed large-scale power grid datasets for load modeling and distributed energy system optimization using statistical and computational methods

## Department of Second Language Acquisition · Research Assistant 2019 – 2020

University of Wisconsin Madison · Madison, WI

- Supported qualitative research on multilingual learning; processed and applied thematic coding to 360+ hours of interview data
- Presented findings at UW Undergraduate Research Symposium

## INDUSTRY EXPERIENCE

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### Yapstone Inc. · Data Engineering Intern Summer 2022

San Francisco, CA

- ↓ **75% manual review**: Built ETL pipeline processing 50K contracts; reduced manual review workload by 75%
- Integrated Snowflake and Looker to enable real-time analytics dashboards for operations team

### Wisconsin Department of Health Services · Data Services Specialist 2021 – 2022

Health Informatics · Madison, WI

- Developed HIPAA-compliant data platform supporting 70K+ patient records
- Improved data reliability by 25–27% and reduced operational costs through pipeline refactoring

## TEACHING

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### Teaching Assistant · CS 200: Software Engineering Aug 2023 – May 2024

University of Wisconsin Madison · 800+ student enrollment · Madison, WI

- Designed programming assignments addressing AI-assisted coding workflows
- Led weekly labs and held 7 hours/week of office hours; received 93% student satisfaction rating

## TECHNICAL SKILLS

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**Languages:** Python, R, Java, SQL, C/C++, JavaScript, .NET (C#)

**ML / AI:** PyTorch, HuggingFace Transformers, scikit-learn, DistributedDataParallel, mT5

**Data & Infra:** Snowflake, Elasticsearch, MS SQL, Google Cloud, Spark, Airflow, Looker

**Methods:** Qualitative study design, IRB protocol, thematic coding, A/B experimentation, NLP, recommender systems

**Research Areas:** Human-centered AI, algorithmic bias, information literacy, social computing, security & privacy, ML systems

## HONORS, AWARDS & SERVICE

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- Student Volunteer — RecSys 2026
- UWSA Fellowship (2024–2025) — 1 of 7 recipients in WI state, awarded for graduate research excellence
- King-Morgridge Scholarship (2019–2023) — full-tuition merit award, University of Wisconsin-Madison
- ASCAM Leadership Excellence Award (2023)
- Dean's Honor List, 7 semesters
- Student Advisory Board Member, UW Madison Computer Science Department
- Secretary, Association of Cameroonians in Madison
- Member, National Society of Black Engineers (NSBE)
- Member, Wisconsin AI Safety Initiative